BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, RAWALPINDI

INTERMEDIATE (11TH CLASS) EXAMINATION, 2021

BIOLOGY

Past Paper 2021

Retyped faithfully from the original board paper — same questions, no changes.

Biology (Objective Type) — Inter. (Part-I)-A-2021 (For all sessions) — Paper Code: 6463

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 6463

Note: NOTE: Write answers to the questions on the objective answer sheet provided. Four possible answers A,B,C and D to each question are given. Which answer you consider correct, fill the corresponding circle A,B,C or D given in front of each question with Marker or pen ink on the answer sheet provided.

Q1. Botulism is caused by:

(A) Salmonella
(B) Campylobacter
(C) Pseudomonas
(D) Clostridium botulinum

Q2. A sheet of muscles which act as floor of chest cavity is called:

(A) Pleura
(B) diaphragm
(C) Intercostal muscles
(D) Lungs

Q3. Starch Sugar hypothesis was proposed by:

(A) H. Van Mohl
(B) Ernst Munch
(C) Ernst Hackel
(D) Loius Pasture

Q4. Attraction among water molecules, which holds the water molecules together, is:

(A) Tension
(B) Cohesion
(C) Adhesion
(D) Transpiration

Q5. Plants having foreign DNA incorporated into their cells are known as:

(A) Transformation
(B) Transgenic
(C) Transgender
(D) Translation

Q6. Heterogenous [sic] group of compounds related to fatty acids are:

(A) Nucleic acids
(B) Protein
(C) Lipids
(D) Carbohydrates

Q7. Which inhibitors have structural resemblance with substances?

(A) Irreversible
(B) Competitive
(C) non competitive
(D) Co-enzyme

Q8. Infoldings of inner membrane of mitochondria are called:

(A) Cristae
(B) Cisternae
(C) F1 particals [sic]
(D) SER

Q9. Enzyme released from the tail of bacteriophage which dissolves cell wall:

(A) Lipase
(B) Pepsin
(C) amylase
(D) Lysozyme

Q10. Reserve food material in cynobacteria [sic] is:

(A) Glycogen
(B) Starch
(C) Lipids
(D) Protein

Q11. Feeding stage of slime molds is called:

(A) Plasmodesmata
(B) Plasmodium
(C) Plasmolysis
(D) Plasma

Q12. Asexual reproduction in penicillium takes place by:

(A) Spores
(B) Budding
(C) Conidia
(D) Fragmentation

Q13. Integumented, indehiscent megasporangium is called:

(A) Ovary
(B) Ovule
(C) Megaspore
(D) microspore

Q14. Larva produced during the life cycle of annelids, is:

(A) Trochophore
(B) Bipinaria
(C) Tad pole
(D) Brachiolaria

Q15. A blue coloured respiratory pigment present in molluscus is:

(A) Haemoglobin
(B) Haemocyanin
(C) Myoglobin
(D) Phycoerthrin

Q16. Second phase of calvin cycle is:

(A) Carbon fixation
(B) Reduction
(C) Regeneration of CO_2 acceptor
(D) Glycolysis

Q17. Chemical formula of chlorophyll 'b' is:

(A) $C_{54}H_{72}O_5N_4Mg$
(B) $C_{54}H_{70}O_4N_5Mg$
(C) $C_{55}H_{72}O_5N_4Mg$
(D) $C_{55}H_{70}O_6N_4Mg$

Biology (Essay Type) — Inter. (Part-I)-A-2021 (For all sessions)

Time Allowed: 2:40 Hours | **Maximum Marks:** 68

Q. Section - I

Q2. Write short answers of any eight parts from the following.

2x8=16

(i) What are enzymes? give one example.

(ii) Differentiate between Anabolic and catabolic reactions.

(iii) Define irreversible inhibitors.

(iv) Give two properties of Enzymes.

(v) Define lichens and give one example.

(vi) What is parasexuality?

(vii) Define kingdom Animalia.

(viii) How asexual reproduction takes place in poriferans? [partially obscured in source scan]

(ix) What are prototheria? Give one example.

(x) Which types of Muscles are found in the body wall of Annelids? [partially obscured in source scan]

(xi) What is Bioenergetics?

(xii) Write complete equation of Lactic acid fermentation.

Q3. Write short answers of any eight parts from the following.

2x8=16

(i) Differentiate between deductive and inductive reasoning with examples.

(ii) Give three basic components of human circulatory system.

(iii) Give importance of mitochondria.

(iv) Define Parasitology.

(v) Write a note on euglenoids.

(vi) What do you know about water molds?

(vii) How brown algae differ from red algae?

(viii) Write down two functions of Golgi complex.

(ix) Write a note on parasitic flagellates.

(x) Why tracheophytes are successful group of land plants?

(xi) Give four functions of blood.

(xii) Differentiate between microphyll and megaphyll leaves.

Q4. Write short answers of any six parts from the following.

2x6=12

(i) Which changes cause inspiration?

(ii) What are Bacilli? Give their types.

(iii) How Venus flytrap cathes [sic] insect?

(iv) Give structure and position of lungs in chest cavity.

(v) How oral cavity helps in selection of food?

(vi) Mention three ways of gaseous exchange in plants.

(vii) Write roles of ventilation and capillary network in respiratory surface.

(viii) Draw labelled diagram of human immunodeficiency virus (HIV).

(ix) What is meant by absorption and assimilation of food?

Q. Section - II — 8x3=24

NOTE: Answer any three questions from the following.

Q5. (a) Discuss how the science of Biology is helping mankind in different ways.

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(b) Discuss the composition of blood Plasma.

4

Q6. (a) What are proteins? Describe primary structure of proteins.

4

(b) Describe different methods of asexual reproduction in fungi.

4

Q7. (a) Write a note on nutrition in bacteria. (b) Write in detail the life cycle of Angiospermic plant.

4+4

Q8. (a) Describe any four viral diseases. (b) Write a note on noncyclic phosphorylation with diagram.

4+4

Q9. (a) Describe the structure and function of Golgi apparatus.

4

(b) Give an account of nutrition in insectivores plants.

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